

LUC MCCUTCHEON

RESEARCH ENGINEER · VLA POST-TRAINING

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lucmccutcheon.home@gmail.com | [GitHub](#) | [LinkedIn](#) | [Google Scholar](#) Research Scientist & PhD Candidate (completing Summer 2026) specialising in Reinforcement Learning for distributed VLA post-training. Co-Founder/CTO with experience deploying full-stack AI solutions and leading technical teams. Expert in JAX/PyTorch for policy optimisation, with published research (ICRA, NeurIPS, IROS) focusing on world models and optimisation algorithms.

RESEARCH SCIENTIST (FIXED-TERM)

Cambridge Consultants — Oct 2025 – Apr 2026

- Led the Reinforcement Learning workstream for the Unitree G1 humanoid, delivering robust policies for dynamic environments.
- Engineered an RL fine-tuning pipeline for VLA models (GR00T 1.6) using PPO/GRPO within custom Isaac Sim environments.
- Designed MuJoCo and Isaac Sim environments supporting curriculum and continual learning to minimise the sim-to-real gap.
- Coupled individual foot control with MPC planners to achieve precise, stability-aware foot placement in 3D space.
- Executed sim-to-real transfer via ONNX to deploy blind stair traversal locomotion on physical hardware.
- Trained cooperative multi-agent RL policies for a safety-critical defence application, using Kubernetes for distributed training.

LEAD RESEARCH SCIENTIST (PART-TIME)

Agile Loop — Oct 2023 – Apr 2025

- Led a team of 10 researchers developing autonomous AI agents, driving technical recruitment and presenting to VC firms.
- Architected agentic pipelines from scratch and steered the internal research strategy, ranging from fundamental deep learning architecture to RL and VLM LoRA fine-tuning.
- Presented our full-stack approach to the Google Cloud team during their workshop for AI agents.

RESEARCH SCIENTIST (PART-TIME)

Agile Loop — Jun 2023 – Oct 2023

- Pioneered RL environments for software and web-apps, using computer vision for icon recognition, and creating systems for distributed multi-task training.
- Developed software and infrastructure allowing hybrid edge/cloud model routing based on task complexity for Lenovo POC.

RESEARCH ASSISTANT (VOLUNTEER)

Connected & Autonomous Vehicle Lab — Jun 2022 – Aug 2022

- Improved exploration using Noisy Neural Networks and tackled partial observability using LSTMs and Deep Q-Learning.

SOFTWARE ENGINEER INTERN

QinetiQ — Jul 2018 – Sep 2018 & Jul 2019 – Sep 2019

- Engineered tactical voice cryptography (C++/Python) and automated red team attacks across two placements, bridging offensive and defensive security operations.

PHD REINFORCEMENT LEARNING (SPONSORED BY VEOLIA NUCLEAR SOLUTIONS)

University of Surrey — 2021 – present

- Designed and implemented time-delay mitigating controller for Veolia Nuclear Solutions' local-remote manipulator (DEXTER).
- Learned a probabilistic World Model to “undelay” RL environments to provide adaptive PD gains in highly stochastic conditions.

- Improved stability analysis through learning a Lyapunov function, using RL to generate counter-examples improving robustness.
- Co-designed and lectured “Intelligent Vehicle Design” university course, teaching planning methods for robotic control.
- Implemented various papers and algorithms from scratch in JAX, such as PCGrad, PPO, ReDo, CBP, VAE, ResNet and many more.
- Implemented numerous custom RL environments, creating complex wrappers for random time delayed environments and local-remote system parallelism.

BSC COMPUTER SCIENCE (HONS) — FIRST CLASS

University of Surrey — 2018 – 2021

- Recipient of the prestigious CyberFirst Scholarship, sponsored by QinetiQ.
- 1st Class Degree, key modules: Artificial Intelligence (92%), Data Structures & Algorithms (86%), Computational Intelligence (83%).
- Dissertation in time-series financial forecasting using an LSTM Neural Network with various financial indicators.
- Key takeaways: CUDA programming, evolutionary algorithms and fundamental mathematics.

Frameworks: JAX, PyTorch, Gymnasium, vLLM, NumPy

Code: Python, C++, Rust, JavaScript, Lua, Bash

Methods: PPO, GRPO, LoRA, VLA, SAC, Rainbow, ResNet

Compute / Infrastructure: Unitree G1, SLURM, GCP, AWS, Docker

Languages: English (Native), French (C1)

First Author

- **Preventing Policy Collapse in Continual Reinforcement Learning** — Under Review. Implemented PPO, Continual Backpropagation (CBP), Recycling Dormant Neurons (ReDo) and a custom optimiser (CPR). Identifies parameter resets as a source of policy collapse in continual RL and improves stability through soft parameter transformations.
- **Neural Lyapunov Function Approximation with Self-Supervised Reinforcement Learning** — ICRA 2025. Efficient probabilistic world model for stable RL under stochastic conditions, jointly learning a SAC control policy, world model and Lyapunov function and exploiting geometric properties for stability analysis.
- **Adaptive PD Control using Deep RL for Local-Remote Teleoperation with Stochastic Time Delays** — IROS 2023. Proposed PMDC, a model-based RL framework using a learned world model ensemble to mitigate stochastic delays up to 290ms.

Co-Author

- **Meta-World+: An Improved, Standardized, RL Benchmark** — ICML 2025 Workshop (Spotlight) & NeurIPS 2025. Implemented optimiser algorithms such as PCGrad and GradNorm for multi-task learning.
- **Exploring In-Context Ensemble with Video-Language Models for Low-Level Workflow Understanding** — NeurIPS 2024 Workshop. Project definition and implementation of Gemini models into a novel in-context ensemble learning framework.
- **Prediction Based Decision Making for Autonomous Highway Driving** — ITSC 2022. Implemented the “Noisy Networks” baseline and part of the Rainbow algorithm.

All implementations are created from scratch using PyTorch or JAX/Flax and trained on a SLURM cluster or Google Cloud, each demonstrating state-of-the-art performance against baselines on challenging benchmarks.

Reviewing

ICLR 2026 | NeurIPS 2025 | TNNLS 2025 | IROS 2025

- Honourable Mention — (xAI) Grokathon 2026
- Bronze — Mathematics Olympiad
- Bronze — British Informatics Olympiad
- Grace Hopper Award — Computer Science
- Foundership Award — Student Enterprise
- NVIDIA Fundamentals of Deep Learning
- Coursera Reinforcement Learning Specialisation
- EPQ in Mathematics
- **Guest Speaker & Panel Discussion (Agents Workshop)** — Google
- **Model-based Reinforcement Learning** — UoS School of Engineering
- **JAX vs PyTorch** — UoS School of Engineering
- **Model Optimisation & Compilers** — Cambridge Consultants
- **Safe Reinforcement Learning** — Cambridge Consultants